KLH BACHUPALLY · DATA STRUCTURES & ALGORITHMS

Medicine Tracker System

Java DSA Implementation

CO1: Search & Sort · CO2: Linked Lists · CO3: Stack/Queue · CO4: HashMap · CO5: Application · CO6: UI

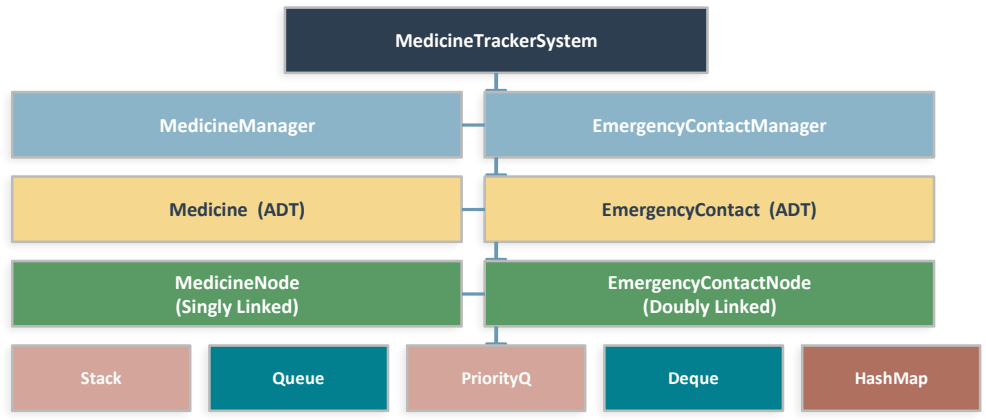
Comprehensive DSA Application for Healthcare Management

Abstract

The Medicine Tracker System is a full-featured Java DSA application implementing core data structures and algorithms for healthcare management. It mirrors the web application's functionality using Java Collections Framework.

The system uses Singly Linked Lists for medicine storage, Doubly Linked Lists for emergency contacts, Stacks for undo/redo, Priority Queues for urgency ranking, and HashMaps for O(1) lookups — all demonstrated with real medicine tracking use cases.

Class Architecture



CO - WISE IMPLEMENTATION

CO1 Searching & Sorting Algorithms

- Linear Search O(n) — sequential traversal of the Singly Linked List node by node with comparison count
- Binary Search O(log n) — sorts ArrayList first, then divide-and-conquer halving; reports comparisons
- Merge Sort O(n log n) — stable, divide-and-conquer sorting by name; uses separate left/right sublists
- Quick Sort O(n log n) avg — pivot-based partition by scheduled time; unstable but cache-friendly
- Bubble Sort O(n^2) & Selection Sort O(n^2) — demo implementations; compareAllSorts() benchmarks all

CO2 ADTs & Linked List Operations

- Medicine ADT — encapsulates name, dosage, scheduledTime, stock, category, notes, refillDate, taken state
- Singly Linked List — MedicineNode(data, next); insert O(n), delete O(n), reverse O(n) in-place
- Doubly Linked List — EmergencyContactNode(data, next, prev); O(1) insert at tail, O(n) delete, bidirectional traversal
- detectDuplicateTimes() — HashMap groups medicines by time slot to find scheduling conflicts

CO3 Stack, Queue & Priority Queue

- Stack<Medicine> — undoStack / redoStack for add-medicine undo/redo operations; O(1) push/pop
- Queue<Medicine> — LinkedList FIFO for scheduled processing order; medicineQueue display
- Circular Queue — array-based simulation with (rear+1)%capacity wraparound for daily schedule
- Deque<Medicine> — ArrayDeque supports offerFirst/offerLast/pollFirst/pollLast at both ends
- PriorityQueue — max-heap by calculated priority: stock level + missed status + category

CO4 HashMap, HashSet & Collections

- HashMap<String, Medicine> medicineMap — O(1) average lookup by lowercase name for markAsTaken/updateStock
- HashMap<String, List<String>> interactionWarnings — drug interaction database; chaining collision demo
- HashMap<String, List<String>> categoryMap — groups medicines by category for the category-wise report
- HashMap<String, List<String>> timeMap — detects duplicate scheduled times across medicines
- Collections.sort(), Collections.swap() — Java Collections Framework used throughout sort algorithms
- HashMap<String, EmergencyContact> contactMap — O(1) emergency contact search by name

CO5 & CO6 Application Features & UI

- Medicine Management — add, view, mark as taken, delete, update stock, priority queue view
- Emergency Contacts — add, search O(1), remove, forward & reverse doubly-linked-list display
- Reports & Analytics — daily summary, adherence rate %, low stock alerts, refill reminders
- Interaction Warnings — HashMap lookup checks active medicines against known drug interactions
- Export Simulation — CSV-format output via printf for medicines_export.csv download simulation
- Console UI — nested menus with switch-case, input validation, banner, formatted table output

DATA STRUCTURE COMPLEXITY AT A GLANCE

Linked List Insert Time: O(n) Space: O(1) Singly LL	**HashMap Lookup** Time: O(1) avg Space: O(1) HashMap	**Binary Search** Time: O(log n) Space: O(1) Sorted Array
Merge Sort Time: O(n log n) Space: O(n) ArrayList	**Priority Queue** Time: O(log n) Space: O(1) Binary Heap	**Stack / Queue** Time: O(1) Space: O(1) Stack/LinkedList

Conclusion & Future Scope

This Java DSA project implements every major data structure — Singly/Doubly Linked Lists, Stack, Queue, Circular Queue, Deque, PriorityQueue, and HashMap — within a practical medicine tracking application, demonstrating real-world algorithmic trade-offs.

Future Scope: Binary Search Tree for O(log n) medicine lookup, Graph-based drug interaction mapping, Trie for prefix-based medicine name search, and File I/O for persistent data storage.

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